



"We're **SoterCare**, a smart care system that monitors, alerts, spots trends and keeps the record, for every resident in a care home."

Ageing faster than care can keep up



WORLDWIDE

1 in 6

people will be **60+** by 2030

2.1 billion people over 60 by 2050, double the number in 2020.

GLOBAL
ELDERLY CARE

6.4M

care home beds in the EU and US

Aged parents move into professional care homes. That is where the paying market is.

CARE WORKERS
SHORTAGE

~13.5M

more care workers needed in OECD countries by 2040



Have ~22.5M

Need ~36M

WHERE WE START
Sri Lanka

The same squeeze, at home: 18% of Sri Lankans are already 60+, and there are about **40,000** nurses against the **88,000** needed. We build and prove SoterCare here first.

Care breaks down, unseen

81%

of care home falls are unwitnessed

83% happen in the resident's own room, where no one is watching.



Help comes late

In one study, 31% of residents who fell waited over an hour on the floor.



Understaffed

Two or three carers per home, many untrained, covering every room at night.



No records kept

Handover runs on memory. **Families get promises, not proof.**

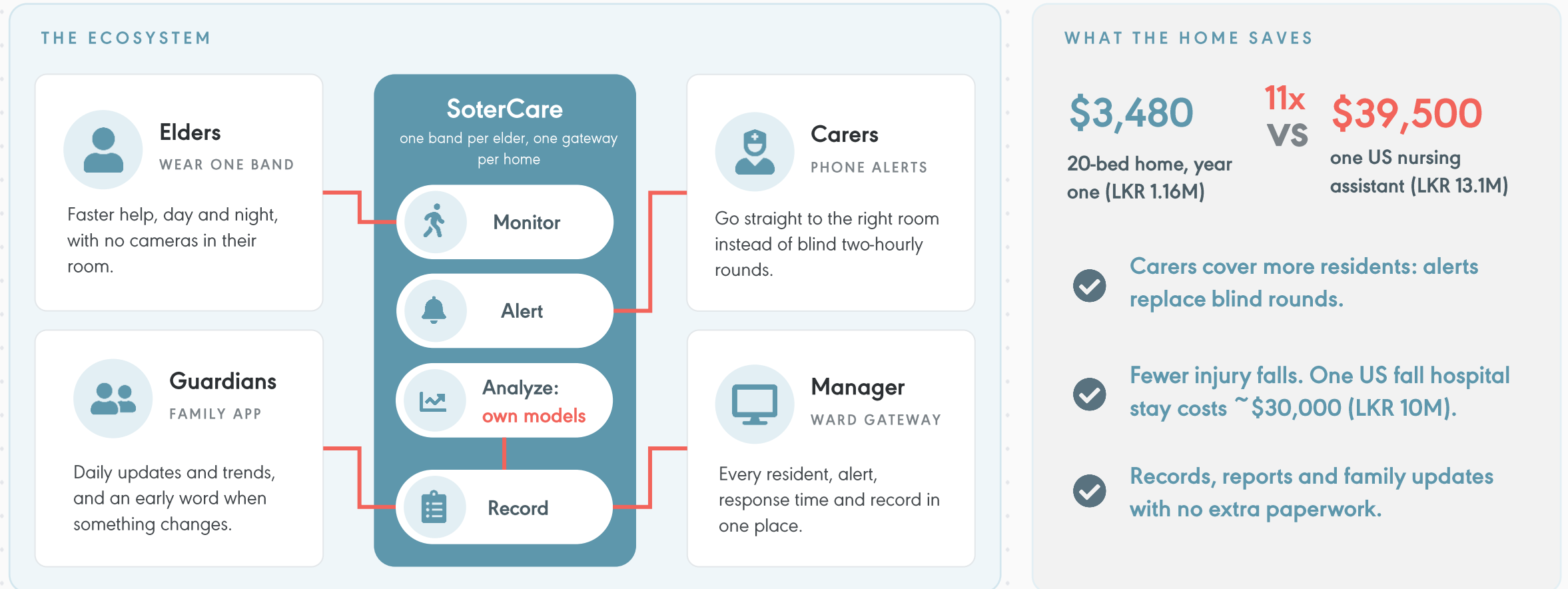
"This is not an eight-hour duty. We must work 24 hours."

Carer, aged care home in Galle

Today's workaround: two-hourly rounds, call bells residents can't press, and CCTV no one watches 24/7.

Connecting elders, carers and families

SoterCare monitors every resident, alerts the right carer in seconds, learns each person's patterns and keeps the home's records, so a care home gives safer care without hiring more staff.



Sensed on the body, analysed at the gateway



Thigh node

Motion, moisture and skin temp sensors; streams readings to the gateway over the home's LAN



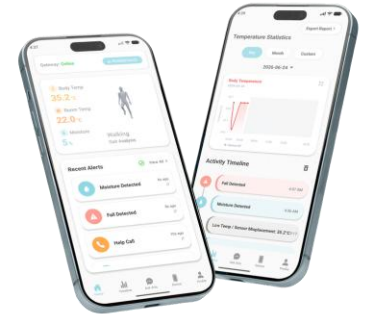
Ward gateway

Runs the gait model and alerts for every bed; 15.6-inch touch screen; USB-C node charging; works offline



Backend Servers

Stores every reading; finds trends in gait, night stand-ups and moisture over weeks



Apps

Carer phones, nurse station, family app; AI writes the shift handover

97.65%

gait classification accuracy, on the gateway (lab data)

Seconds

from a stand-up attempt to the carer's phone, no internet needed

24/7

continuous recording. No cameras, ever

ROADMAP

- Now: TRL 6, working prototype
- Months 1 to 8: full build
- Months 9 to 10: testing, then release

Three global shifts opened the window



IoT hardware got cheap

THEN

A clinical gait sensor cost about \$2,400 (LKR 797,000) per unit.

NOW

A 6-axis motion sensor (IMU) now costs about \$1.40 (LKR 460). One gateway runs the gait model for a whole home.



Regulators want proof

THEN

Care quality was judged on inspection-day paperwork.

NOW

Japan pays care homes extra for using monitoring sensors (2024). Australia tightened its care standards (Nov 2025).



Privacy law vs cameras

THEN

Cameras in bedrooms went unquestioned.

NOW

Under GDPR, UK regulators expect consent, a privacy assessment and less intrusive options first. Camera-free wins.

Prove it in Sri Lanka, sell it to the world

TAM · global care beds paying ~\$400+ a month

~12M beds

\$1.0B (LKR 335B) / yr
+ \$670M (LKR 222B) hardware

OECD ~10.3M, China ~1.3M, Taiwan and Hong Kong ~0.2M

SAM · Sri Lanka's fee-paying care homes

~1,250 beds

\$105k (LKR 34.9M) / yr
+ \$100k (LKR 33M) hardware

SOM by year 3 · 20 care homes

~500 beds

\$42k (LKR 13.9M) / yr
+ \$40k (LKR 13M) hardware

THE FORMULA

**Beds × \$7 (LKR 2,300) a month × 12 +
\$40 a bed + \$1,000 a home, once**



FIRST TARGET

**Registered, fee-paying care homes in
Western Province**

NEXT

**The US: 2.9M beds in 46,900 homes, worth
\$244M (LKR 81B) a year. Then Europe.**

One wearable for alerts, moisture and trends

	Bed-exit sensors (Alert-iT, DAZA)	TENA SmartCare Change Indicator	Camera AI (Kepler Night Nurse)	Radar (Vayyar Care)	SoterCare
Alerts when a resident tries to stand up alone	✓	✗	✓	⦿	✓
Detects a wet pad	✗	✓	✗	✗	✓
Records gait and activity continuously	✗	⦿	⦿	⦿	✓
ML trends over weeks, per resident	✗	✗	⦿	⦿	✓
No cameras in the bedroom	✓	✓	✗	✓	✓
Keeps alerting without internet	✓	⦿	✓	⦿	✓
Typical price	€170 to €720 per bed	Sensor plus strips every 12 weeks	Ceiling camera per room, enterprise	~\$240 per device + monthly fee	\$40 a bed, \$1,000 gateway, \$7 a month

⦿ Partly: TENA logs pad saturation only; camera AI and radar log room activity, not per-resident gait; cloud-linked alerts depend on the network.

Moat: the only single system doing all six, plus a growing per-resident gait dataset our trend models learn from.

Sources: Abilia Alert-iT; eastin.eu (bed-exit alarm RRP); DAZA OptiSCAN; TENA SmartCare (tena.co.uk); Kepler Vision / MOBOTIX NurseAssist; Vayyar Care reviews, 2026.

Healthy hardware margins, then a fee per bed

THIGH NODE

\$40 (LKR 13,300)

one per resident

Our cost **\$19 (LKR 6,300)**

Gross margin **52%** (\$21)

WARD GATEWAY

\$1,000 (LKR 332k)

one per home, 15.6-inch touch screen

Our cost **\$500 (LKR 166,000)**

Gross margin **50%** (\$500)

SUBSCRIPTION

\$7 (LKR 2,300)

per bed per month

Cost to serve **~\$2 (LKR 660)**

Gross margin **~70%** (\$5)

Per home: a 25-bed home pays \$2,000 (LKR 664,000) once for 1 gateway and 25 nodes (~\$1,025 our margin), then \$84 (LKR 27,900) a bed a year.

All of it is about **3%** of a typical LKR 100,000 monthly room fee, with hardware spread over 3 years.

TRL 6 today, TRL 7 next with NIA

TRL 5 DONE

Components validated in a relevant environment



TRL 6 WE ARE HERE

Prototype system demonstrated in a relevant environment



TRL 7 NEXT, WITH NIA

Pilot in real care homes through the NIA voucher programme

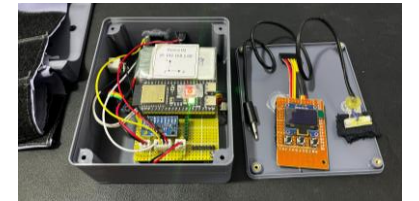


TRL 8 TO 9 THEN

Production release and first paying homes



All in [SoterCare.com/news](https://www.sotercare.com/news)



WORKING WITH



INFORMATICS
INSTITUTE OF
TECHNOLOGY



NIA
NATIONAL
INNOVATION
AGENCY
SRI LANKA
National Innovation Agency (NIA)



We build the whole stack ourselves



Daham Dissanayake

Founder · IOT and ML

Built the node, gateway and gait model.



Sanjula Herath

Co-founder · Backend and AI

Built the cloud backend, mobile app and AI agents.



Komudi Dhara

Co-founder · Design and Docs

Designed the apps and leads documentation.



Banu Athuraliya

EXTERNAL ADVISOR

Visiting Lecturer/SDGP
Module Leader
Digital Consultant
Informatics Institute of
Technology (IIT)



Dr. P.A.D.M. Senarachchi

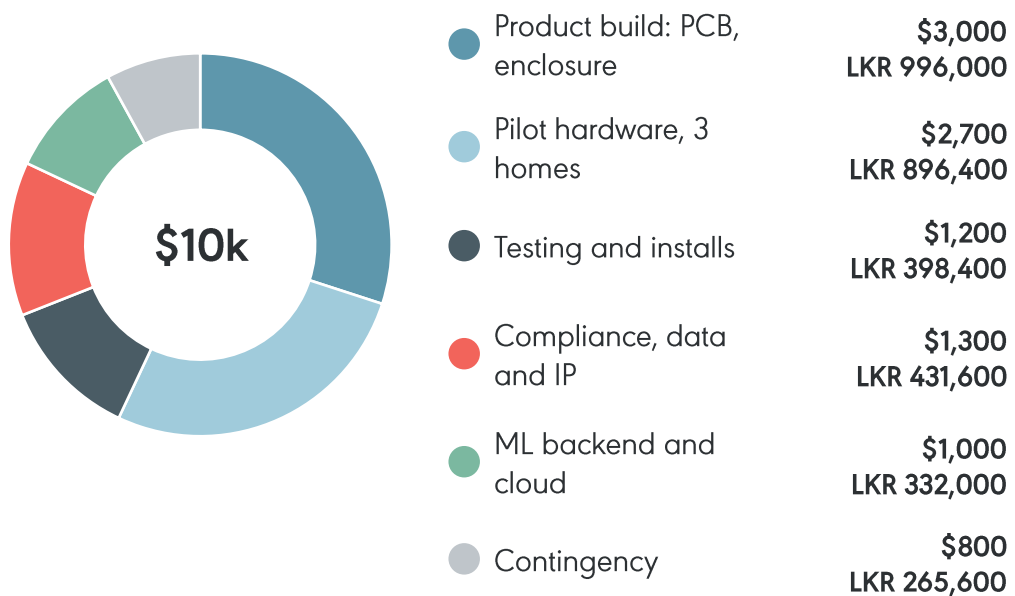
MEDICAL EXPERT

Accident and Emergency Unit
National Hospital, Kandy.
Trauma and Stroke
Rehabilitation Hospital,
Digana

Why us: SoterCare started with caring for our own grandparents, and every layer you've seen, from firmware to ML, is ours.

Raising \$10,000 (LKR 3.32M)

USE OF FUNDS · 12 MONTHS



Pilot hardware at cost: 3 gateways and 60 nodes, about \$2,640 (LKR 876,500). NIA voucher covers part of compliance and the field trial.

Evidence at the next raise: real-world stand-up alerts, false alarms, response times, early trend data, and a paying home.

MILESTONES

Months 1 to 8

Build the full system

Production node and gateway, LAN alerting, moisture sensing, ML trend backend, carer and family apps

Months 9 to 10

Test in 3 homes, 60 beds

Targets: 80% wear time, under 1 false alarm per bed a week, response under 3 minutes

Month 11 onward

Release and first paying homes

Pilot homes convert at a pre-agreed price; outcome report ready for a seed round



THIS IS HOW WE BUILD

Smarter care for every elder!



sotercare.com

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linkedin.com/company/sotercare



Market sizing: the working

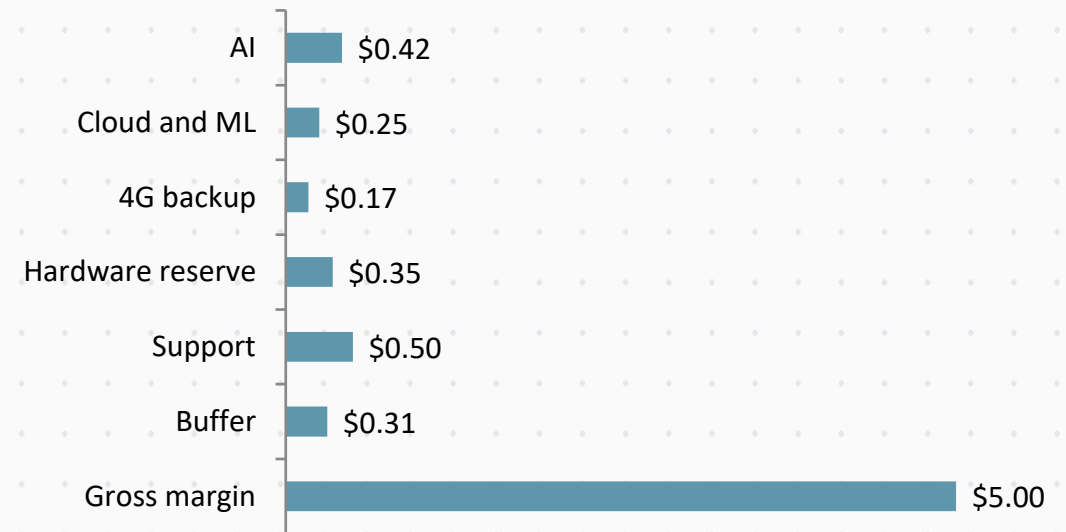
Market and segment	Beds	Fee / bed / month	Subscription / year	Hardware, once
TAM: global care beds paying ~\$400+ a month	~12M	\$7 (LKR 2,300)	\$1.0B (LKR 335B)	\$670M (LKR 222B)
of which OECD countries (OECD Health at a Glance 2025)	~10.3M	\$7 (LKR 2,300)	\$865M (LKR 287B)	\$578M (LKR 192B)
of which China, beds paying 2,900+ yuan a month (estimate)	~1.3M	\$7 (LKR 2,300)	\$109M (LKR 36B)	\$73M (LKR 24B)
of which Taiwan (~110,000) and Hong Kong (~76,000)	~0.19M	\$7 (LKR 2,300)	\$16M (LKR 5.2B)	\$10M (LKR 3.5B)
SAM: Sri Lanka fee-paying care homes (~50 × 25 beds)	~1,250	\$7 (LKR 2,300)	\$105k (LKR 34.9M)	\$100k (LKR 33.2M)
SOM, year 3: 20 care homes (~25 beds each)	~500	\$7 (LKR 2,300)	\$42k (LKR 13.9M)	\$40k (LKR 13.3M)
Next market: US care homes (CDC NCHS 2022)	2.9M	\$7 (LKR 2,300)	\$244M (LKR 81B)	\$163M (LKR 54B)

TAM = OECD (252M people aged 65+ × ~41 long-term care beds per 1,000) + China care beds paying 2,900+ yuan (\$400+) a month (team estimate: ~25% of 5.2M institutional beds) + Taiwan (MOHW) + Hong Kong (SWD). Fees in these markets run well above \$400 a month; in Sri Lanka, premium homes clear it (Royal Nursing Home's Mount Lavinia rooms start at about \$560), while a typical fee-paying home charges about LKR 100,000. Hardware = \$40 (LKR 13,300) per bed + one \$1,000 (LKR 332,000) gateway per home, assuming ~62 beds per home in TAM markets and ~25 in Sri Lanka. Priced at our \$7 Sri Lankan fee, so conservative. Totals rounded. USD to LKR at 332.

Hardware pricing, margins and cost to serve

Item	Our cost	Price	Margin
Thigh node	\$19 (LKR 6,300)	\$40 (LKR 13,300)	52%
Ward gateway, 15.6-inch touch	\$500 (LKR 166,000)	\$1,000 (LKR 332,000)	50%
25-bed home kit	\$975 (LKR 323,700)	\$2,000 (LKR 664,000)	51%
Per bed, 25-bed home	\$39 (LKR 12,900)	\$80 (LKR 26,600)	51%
Per bed, 60-bed home	\$27 (LKR 9,100)	\$57 (LKR 18,800)	52%

Where \$7 (LKR 2,300) per bed per month goes

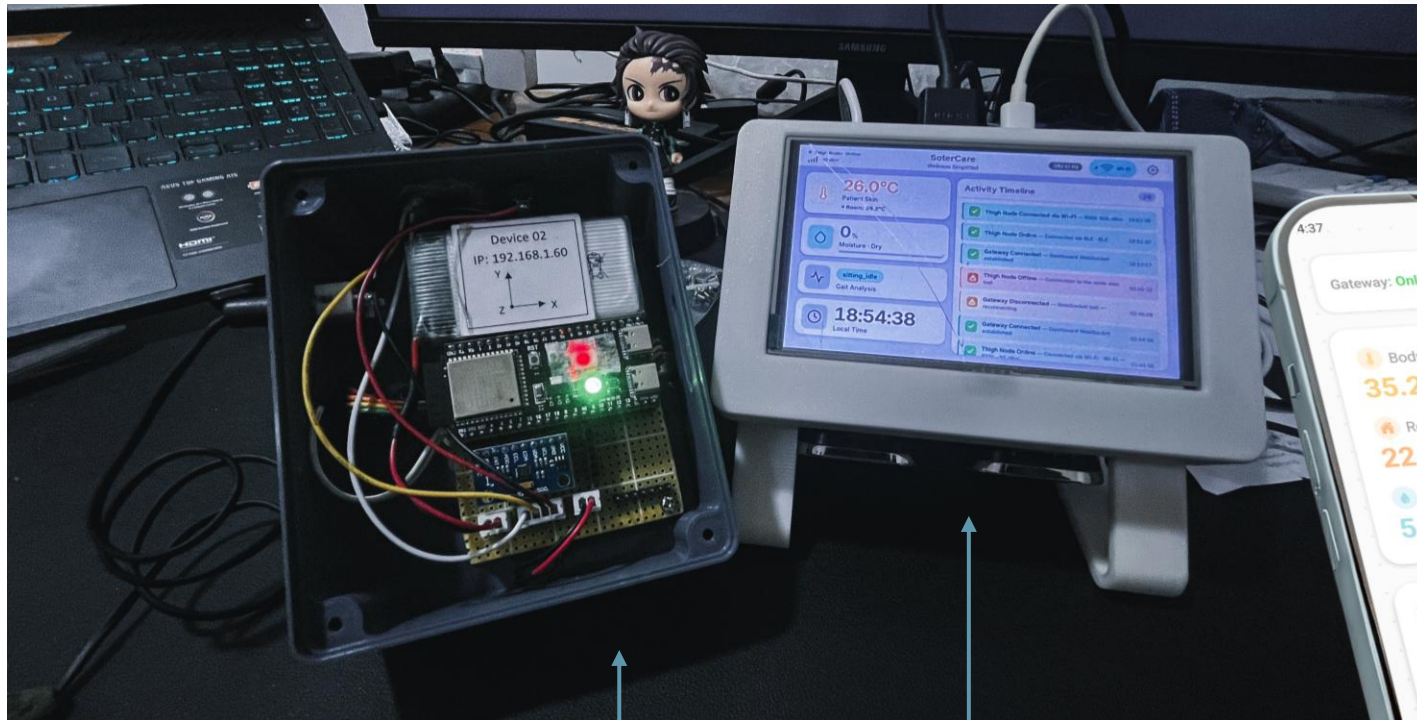


Gateway cost: Compute Module 5 \$100, 15.6-inch touch panel \$125, carrier board with 8 USB-C charging ports \$45, LTE \$22, battery backup \$28, enclosure and stand \$48, power supply \$15, speaker and cables \$17, assembly and test \$25 = \$425, +3% yield and +15% freight and duty = ~\$500. Node landed at 1,000 units. Excludes one-time tooling (\$5k to \$12k). Estimates, to be replaced with supplier quotes.

Known risks, and how we handle them

Risk	Mitigation
Residents remove or refuse the wearable, especially with dementia	Soft thigh strap under clothing; measure wear time per resident; target 80%
Charging becomes a burden on scarce staff	7-day battery target; 8 USB-C charging ports on the gateway; low-battery alerts
False stand-up alerts cause alarm fatigue	Report false alarms per bed a week, not lab accuracy; carer confirm or dismiss retrains the model
Lab gait accuracy does not transfer to real homes	Testing phase logs every stand-up attempt against every alert
Home Wi-Fi or LAN is unreliable	Site survey before install; gateway buffers data and alerts locally
Medical device classification (NMRA)	Position as safety monitoring and records, not diagnosis; ask NMRA for categorisation early
Homes with low budgets can't pay upfront	Start with fee-paying homes; hardware-as-a-service option; CSR sponsors for charity homes

Current working prototype



Mobile App



Smart Thigh Node (Prototype)

ESP32-S3 · IMU · IR temp · moisture clip
· haptic feedback · SOS



Edge Gateway

Raspberry Pi 5 · local ML · 5"
touchscreen dashboard · voice +
buzzer

Final Product Designs



The Thigh Node



The Ward Gateway

Proactive Gait Analysis Model

A custom classifier on the gateway that recognises stand-up attempts in real time, paired with a private haptic warning.



SoterCare Studio - custom data-capture tool (50 Hz IMU)

97.65%

testing accuracy

Keras NN + K-means anomaly detection,
deployed as an AARCH64 library on the
Pi 5

Five movement states classified in real time

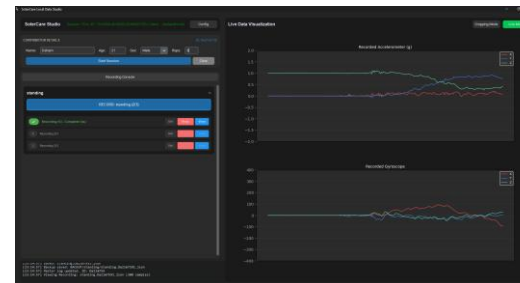
Walking

Standing idle

Sitting idle

Standing up

Sitting down



A Multi-Agent AI Brain

Retrieval-Augmented Generation - three asynchronous pipelines that reason with medical context instead of dumping raw data.



Safety Guard

Real-time · no LLM · Agent 1

Polls sensor readings every 10 seconds, checks safety thresholds, writes alerts to the DB, and pushes WebSocket notifications. No LLM.

db_poller · nurse_agent



ARIA Chat bot

Chat · RAG · Agent 2

Provides a RAG chatbot answering questions about the patient and their behavior, using health summaries for Agent 3.

health_agent · FAISS RAG



Clinical Report Agent

6h · persistent · Agent 3

Fetches 6-hour sensor windows, retrieves historical context from Agent 2, and generates structured health reports via LLM.

report_rag · ChromaDB